# SAMPLE ENVIRONMENTAL PRODUCT DECLARATION

Ready-mixed concrete, declared unit 1 m3, cradle to gate (modules A1-A3)

| | |
|---|---|
| **Declared by** | Sample Ready-Mix Co. (fictional company) |
| **Product** | Ready-mixed concrete, mix code MIX-A-PC380 |
| **Mix** | Mix A (contractor's declared baseline, standard spec) |
| **Role in the demo** | Baseline mix (standard spec) |
| **Cement** | Portland cement Type I/II (CEM I equivalent), 380 kg per m3 [1] |
| **Declared unit** | 1 m3 of ready-mixed concrete at the plant gate |
| **Document number** | SAMPLE-EPD-A (sample, not registered with any program operator) |
| **Status** | Sample for software testing. Not third-party verified. Not for procurement or reporting. |

## Results per declared unit (1 m3)

| Indicator | Value | Unit | Source |
|---|---|---|---|
| GWP of cement (A1-A3) | 370.5 | kg CO2e/m3 | [1] |
| GWP of other constituents | 20 to 40 | kg CO2e/m3 | [2] |
| **GWP-total (A1-A3)** | **390 to 411** | **kg CO2e/m3** | **[3]** |

### Result in one line

**GWP-total, modules A1-A3: 390 to 411 kg CO2e per m3 of concrete.**

The result is a range, not a single number: the cement share is computed from a published EPD and the rest is an assumed band.

### Notes

[1] Cement: 380 kg x 974.92 kg/t (published cement EPD, NRMCA EPD No. 20243). That EPD gives 974.92 kg CO2e per tonne for Type I/II cement on page 5. Cement content 380 kg per m3 is an assumption (typical over-designed C30/37 mix).

[2] Other constituents: assumption (aggregates, water, admixture, batching; not modeled).

[3] GWP-total = [1] + [2], in whole kg as declared on chain by the contractor (baseline) and the supplier (delivered mix).

The cement EPD follows a US product category rule (NRMCA program), not EN 15804. It is used only to anchor the sample range.

SAMPLE - NOT A REAL DOCUMENT

## Why this document is a sample

PourProof records each concrete pour on chain and lets software read the delivery ticket and the EPD. To show that step without using a real company's paperwork, this project made its own documents. The supplier, plant, document number and mix code are invented.

The numbers are not invented. The cement share comes from a published EPD for a Turkish cement (NRMCA EPD No. 20243). The cement content per m3 and the band for aggregates, water, admixture and batching are assumptions, labelled as such here and in data/scenario/inputs.json.

On chain, the sha256 of this PDF file is recorded in a signed transaction. Change one byte and the hash no longer matches.

Do not use these values for design, procurement, carbon reporting or any claim. Real projects need the EPD of the actual mix from its producer, verified under a recognized program.

### Calculation

Cement: 380 kg/m3 x 974.92 kg CO2e/t = 370.5 kg CO2e/m3

Other constituents: 20 to 40 kg CO2e/m3 (assumption)

GWP-total (A1-A3): 390 to 411 kg CO2e/m3

### Source

Published cement EPD: NRMCA EPD No. 20243, valid 26 Nov 2024 to 26 Nov 2029, declared unit 1 metric ton of cement.
Link: data/sample-docs/README.md

SAMPLE - NOT A REAL DOCUMENT